Lucius Gibbons

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Education

- B.S. Mechanical Engineering – UC Berkeley, Regents' and Chancellor's Scholar — May 2028
- A.S. Physics – Irvine Valley College, GPA: 4.0 — May 2026
- Coursework: SolidWorks, Mechanics, Statics, Materials Science, Inventor Professional, Onshape, Hand Drafting

Work Experience

Incoming Mechanical Engineering Intern – Dune Industries, Santa Ana, CA — May 2026 - August 2026

Assembly Tech/Field Tech/Engineering Intern – Bambeck Systems, Santa Ana, CA — January 2025 – August 2025
- Resolved a 2-year critical system failure onsite, redesigning desiccant control process to reduce downtime by 100%
- Designed and manufactured a custom insulator solution in-house, decreasing error throws from overheating by 18%
- Revised and implemented in-factory assembly processes, reducing build times for 500+ component analyzers by 3 days
- Served as Lead Field Servicing Engineer during a staffing shortage, completing 4 solo service trips ahead of schedule

Research Intern – California State University, Fullerton — June 2024 – August 2024
- Built and trained an LSTM Deep Learning model to detect wildfire presence from smoke sensors with 99% accuracy
- Reduced dataset compilation/processing times from >1 hr to 3 minutes by automating file retrieval process

Elementary School Tutor – Step Up Tutoring, Irvine, CA — August 2023 – June 2024
Woodworking Assistant – CocoDealers, Los Angeles, CA — July 2022 – September 2023
Student Worker/Shift Lead – Greenway Arts Alliance, Los Angeles, CA — June 2022 – August 2023

Projects

IVC Baja – Founder, Project Manager — July 2024 – July 2025
- Recruited 18 students and 2 faculty to design and manufacture an off-road vehicle for the Baja SAE 2025 Season
- Architected chassis design ground-up, reducing complex bends by 100% & optimizing for available bend dies
- Fundraised $4000 through local outreach, organizing fundraisers, and oversight of dedicated business team
- Hosted workshops of 30+ students on DFMA, CAD management in Bild, and SolidWorks design best practices

5DoF Robotic Arm – Saddleback Mars Rover Team — November 2025 – February 2026
- Designed new robotic arm in SolidWorks, doubling load capacity and weighing 5kg less compared to previous iteration
- Reduced CNC operations on 20+ custom parts using DFM principles, saving $8000+ and cutting lead times by weeks
- Sourced 200+ COTS components, calculating torque, gear ratios, and power draw for optimal motor selection
- Redesigned custom linear actuator and belt drive to shift 9kg off-arm, increasing load capacity by 6kg and speed by 4x

Soil Sampler and Life Detection System – Saddleback Mars Rover Team — September 2025 – December 2025
- Spearheaded complete mechanical design of rover's science payload, developing sketches into a field-tested assembly
- Devised a passive stabilizing mechanism for drill platform via gas springs, ensuring drilling stability on uneven terrain
- Modelled and 3D printed a monocoque light-sealed optical analysis chamber, cutting potential light-leak failure points
- Packaged 12 actuators and a fluidics loop into rover chassis, with hot-swappable vibration dampening mount solutions

Monopiece TPU Tirewheel – Saddleback Mars Rover Team — August 2025 – September 2025
- Created a 3D printed wheel with 0.4x mass and 3x deflection vs old COTS wheel, resulting in rover's first stair climb
- Designed for Mars conditions, with an open 92% air compressible gyroid lattice body for 0 maintenance/deflation
- Optimized fillets, chamfers, and overhangs to create reliable, printable geometry, with all 4 wheels printing first-try

Skills

Software: SolidWorks (CSWA), Autodesk Inventor (Autodesk Certified), Autodesk Fusion, Onshape, Python, Excel
Technical: 3D Printing, Soldering, Woodworking, Manual Milling, CNC Router, Vernier Calipers, Hacksaw
Design: DFMA (CNC, 3D printing, Injection Molding, Sheet Metal), Rapid Prototyping, Electronics Selection

To see my projects, view my portfolio at luciusgibbons.com